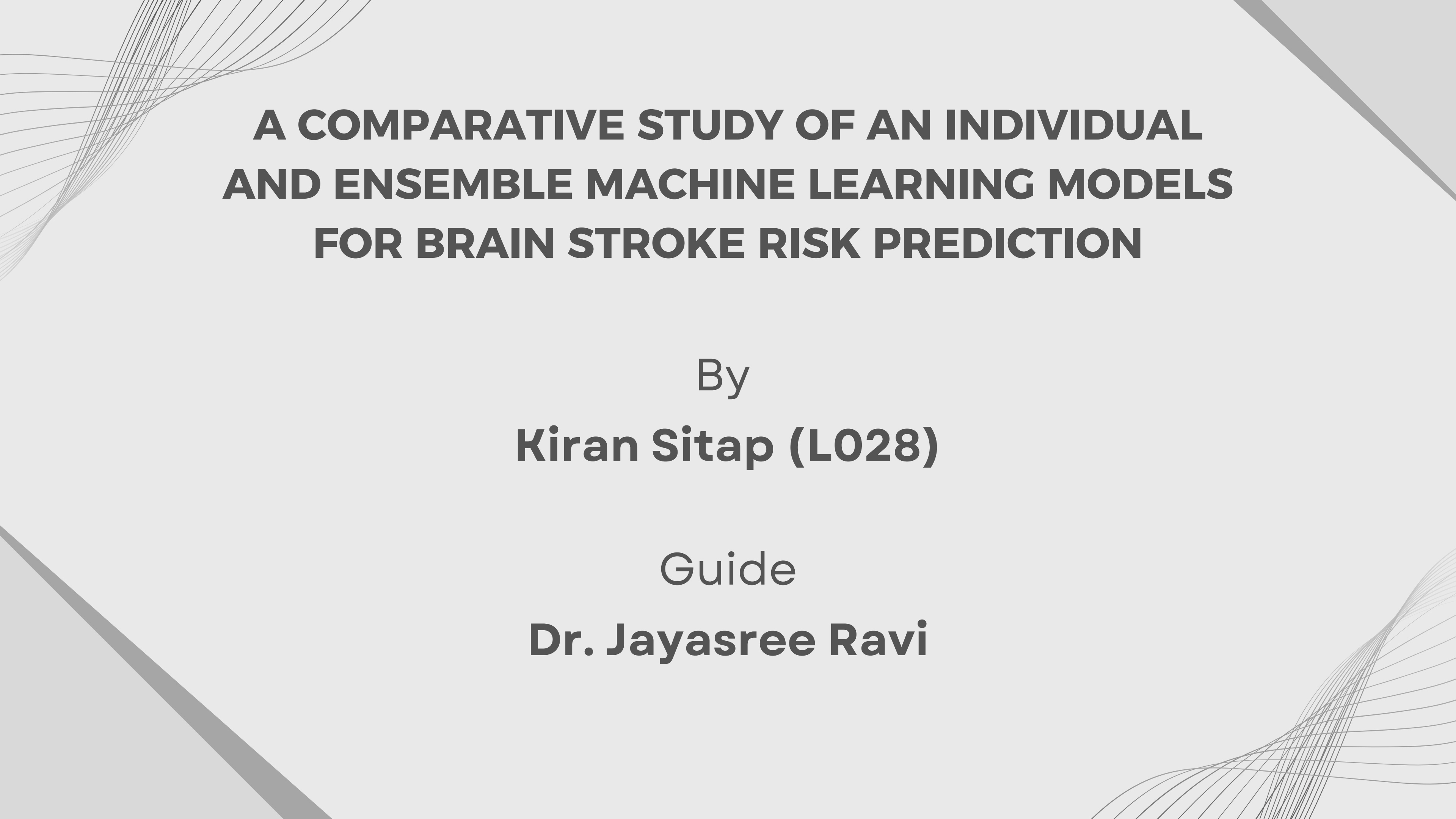




A COMPARATIVE STUDY OF AN INDIVIDUAL AND ENSEMBLE MACHINE LEARNING MODELS FOR BRAIN STROKE RISK PREDICTION

By

Kiran Sitap (L028)

Guide

Dr. Jayasree Ravi



INTRODUCTION

- Stroke is a serious and life-threatening condition where early risk prediction can improve patient outcomes.
- Stroke risk prediction is challenging due to the involvement of multiple clinical and symptom-based factors.
- Machine learning models can analyze these factors together to understand and predict stroke risk effectively.
- This project compares individual and ensemble machine learning models to study their performance in stroke risk prediction.

PROBLEM STATEMENT

- Stroke risk prediction involves multiple clinical conditions and symptoms, making it difficult to assess using traditional methods.
- Single machine learning models may not always provide reliable or consistent predictions.
- There is a need to analyze and compare different machine learning approaches to understand their effectiveness in stroke risk prediction.

Problem:

How can individual and ensemble machine learning models be evaluated and compared for effective stroke risk prediction using clinical and symptom-based data?



OBJECTIVES

- To analyze stroke risk using clinical and symptom-based data.
- To implement individual machine learning models for stroke risk prediction.
- To apply an ensemble approach using a Voting Classifier.
- To evaluate and compare model performance using appropriate evaluation metrics.



METHODOLOGY

Dataset

- Stroke risk dataset with 35,000 patient records.
- Includes demographic, clinical, and symptom-based features.

Preprocessing

- Dataset exploration and feature inspection.
- One-hot encoding for categorical features.
- Standardization of numerical features.
- Removal of target-derived and near-diagnostic features.
- Train-test split for model evaluation.



METHODOLOGY

Models Used

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

Ensemble Model

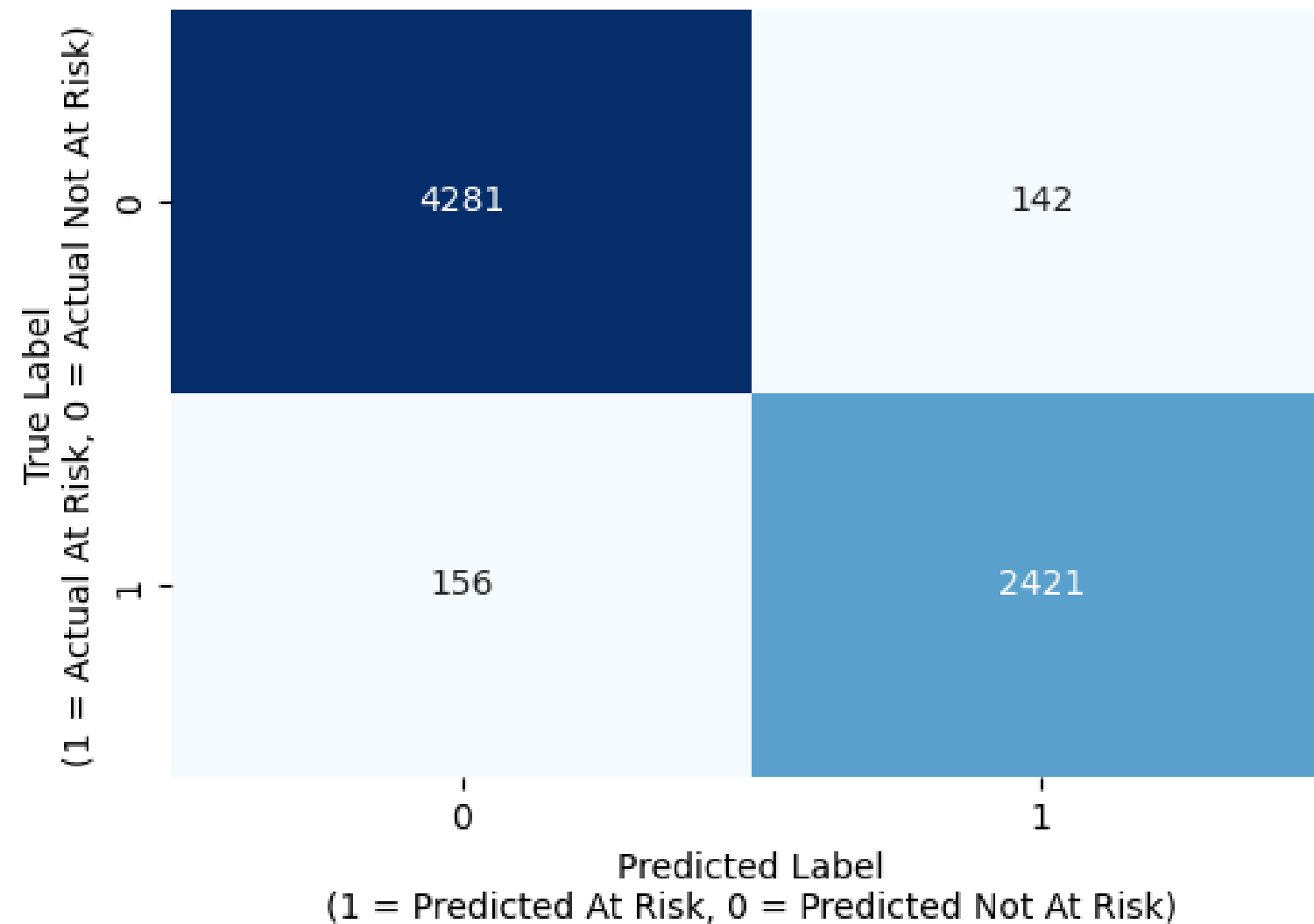
- Voting Classifier combining the above models using soft voting

Evaluation Metrics

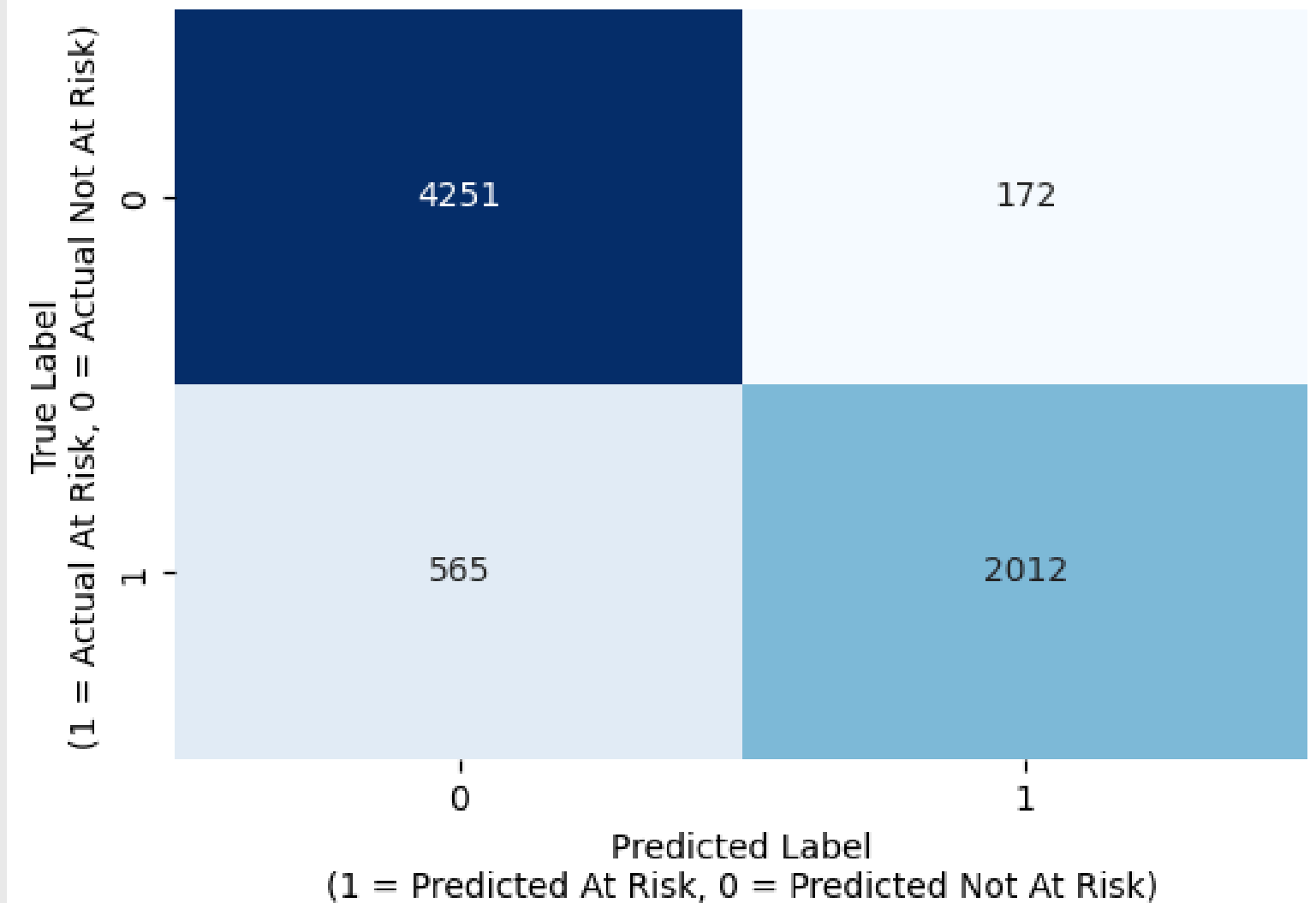
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

MODEL COMPARISON

Logistic Regression - Confusion Matrix

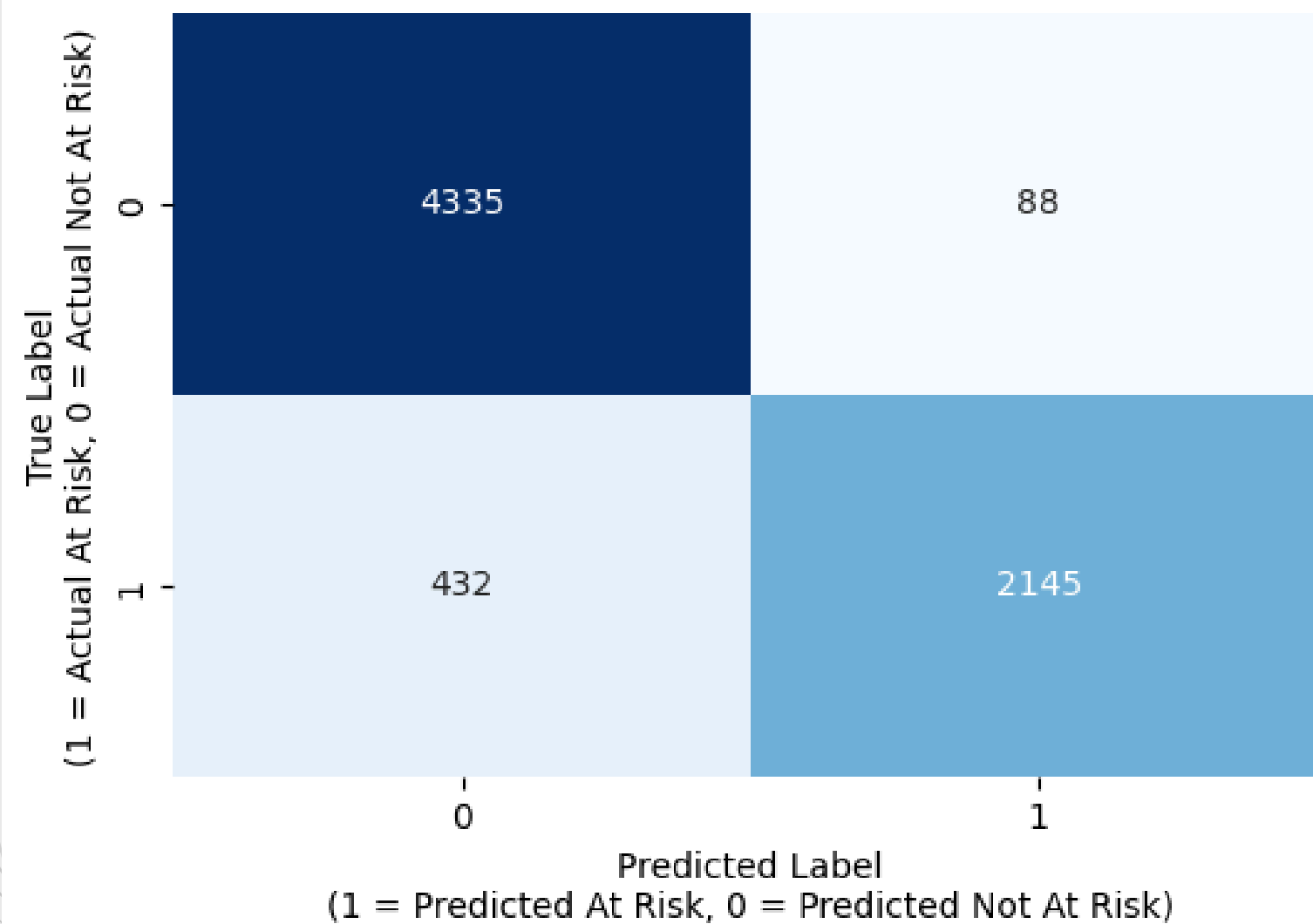


Decision Tree - Confusion Matrix

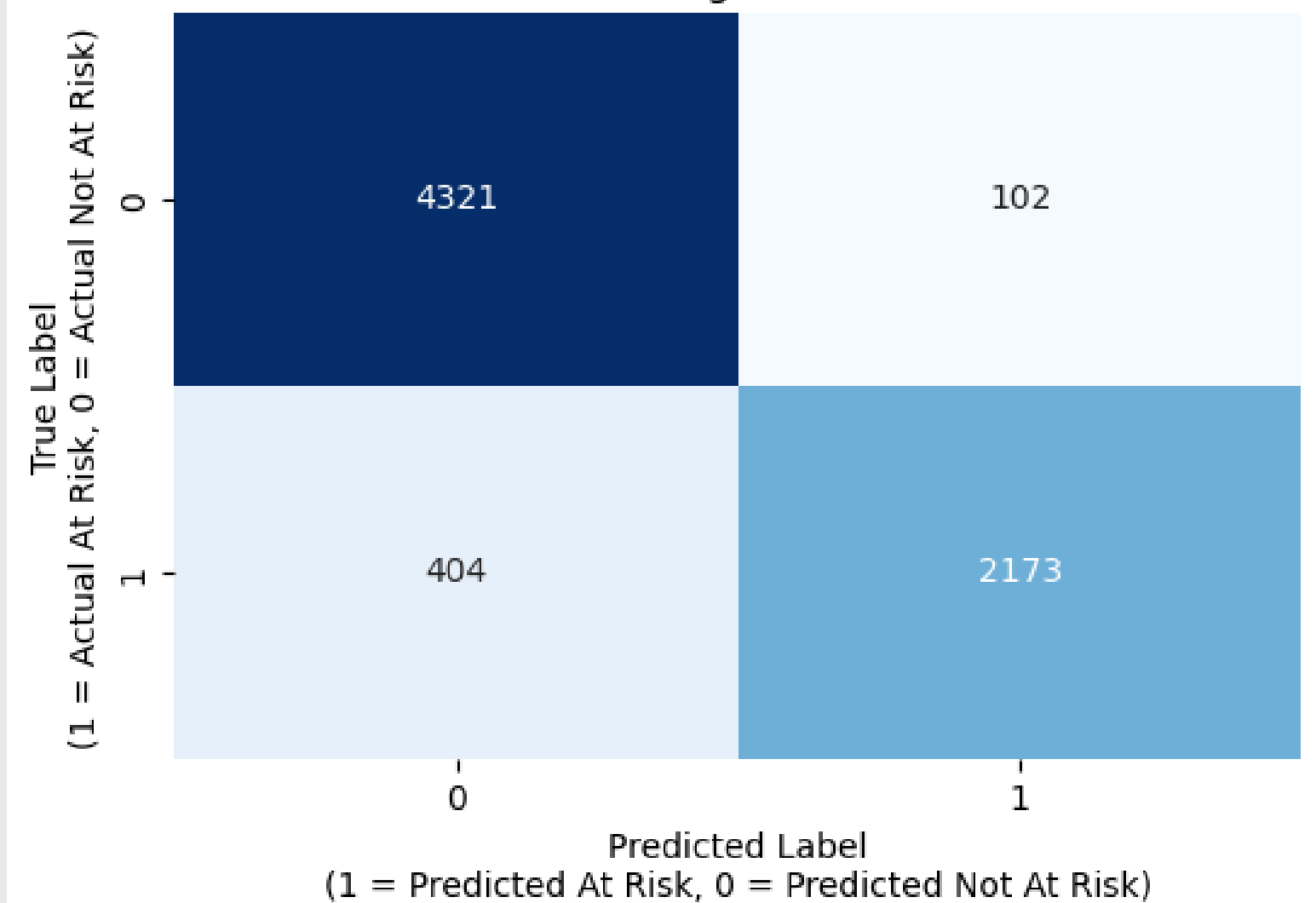


MODEL COMPARISON

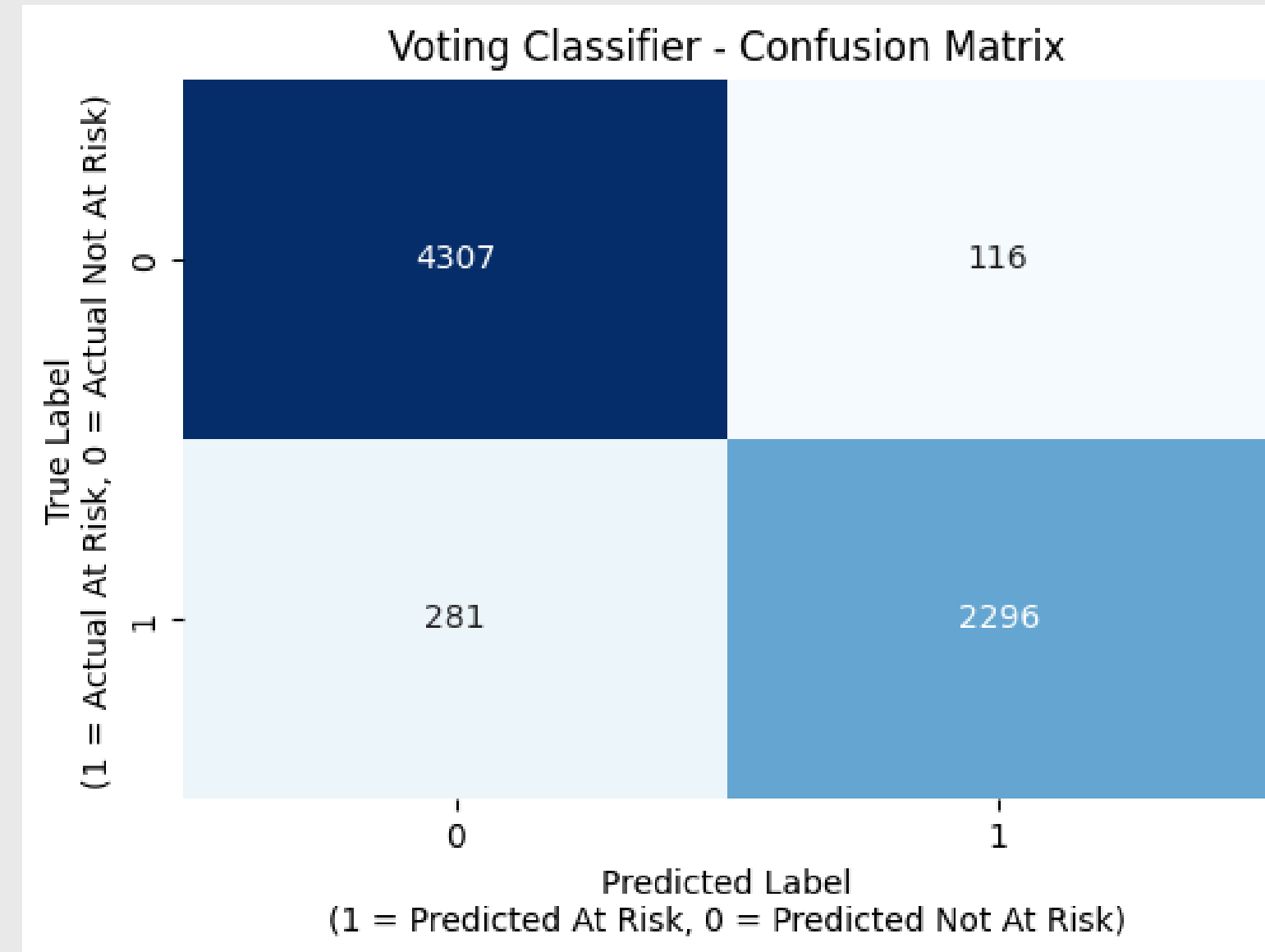
Random Forest - Confusion Matrix



Gradient Boosting - Confusion Matrix

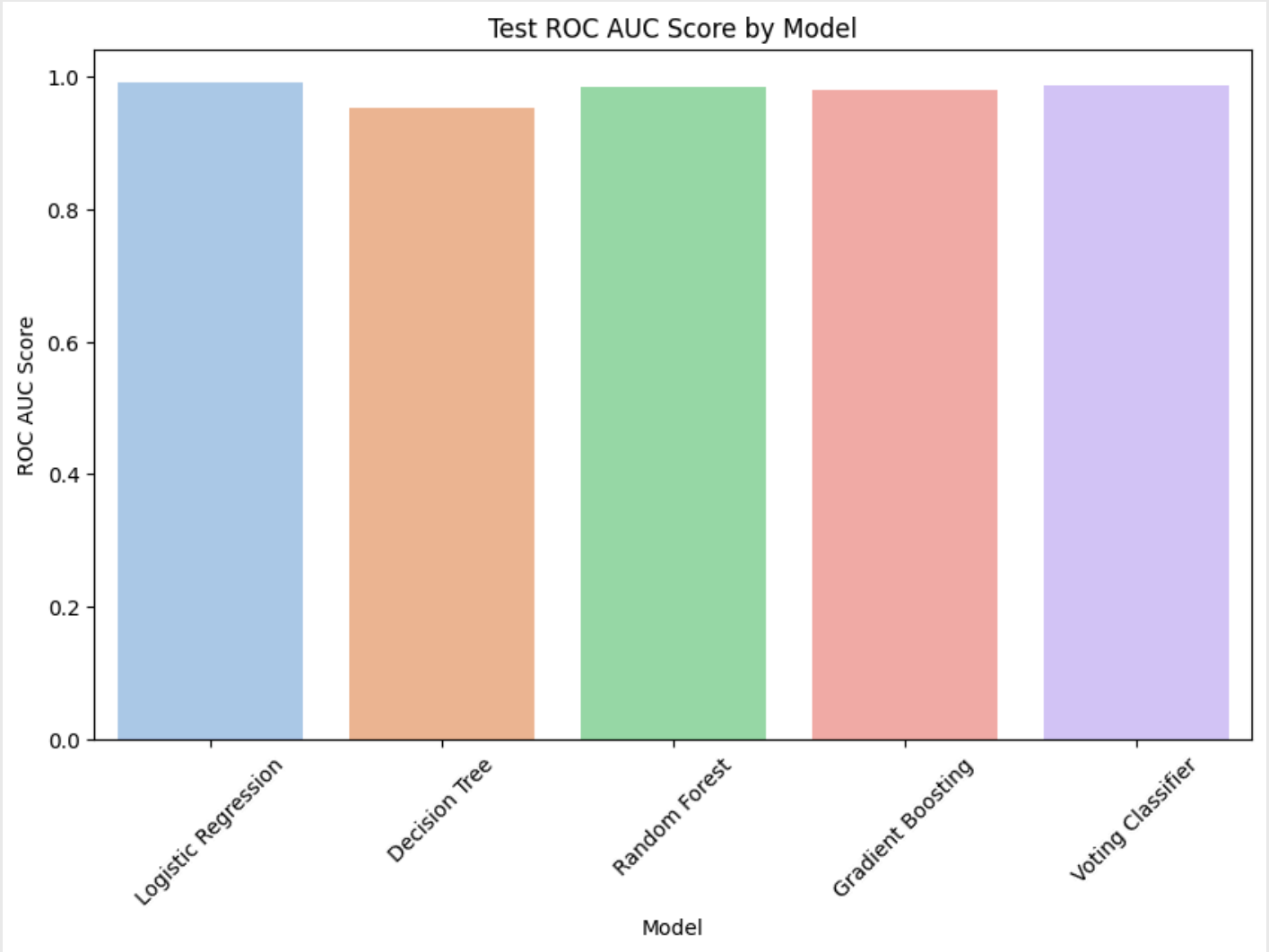


MODEL COMPARISON



Summary of Model Performance:

	Model	Test Accuracy	Test Precision	Test Recall	Test F1 Score	Test ROC AUC
0	Logistic Regression	0.957429	0.944596	0.939464	0.942023	0.992253
4	Voting Classifier	0.943286	0.951907	0.890958	0.920425	0.987832
2	Random Forest	0.925714	0.960591	0.832363	0.891892	0.984592
3	Gradient Boosting	0.927714	0.955165	0.843229	0.895713	0.981563
1	Decision Tree	0.894714	0.921245	0.780753	0.845201	0.952603



PREDICTION

Case 1 - At Risk

Age	<input type="range" value="45"/>	45
Gender	Female	▼
Chest Pain	0	▼
High BP	1	▼
Short Breath	0	▼
Fatigue/We...	1	▼
Dizziness	0	▼
Edema/Sw...	1	▼
Neck/Jaw ...	0	▼
Persistent ...	0	▼
Nausea/Vo...	0	▼
Chest Disc...	1	▼
Cold Hand...	0	▼
Sleep Apnea	0	▼

Predict Stroke Risk

⚠ Prediction: Patient is at HIGH RISK of Stroke
Stroke Risk Probability: 0.9187
Stroke Risk Percentage: 91.87 %

Age: 45 years

Gender: Female

Symptoms:

- High blood pressure
- Fatigue
- Edema
- Chest discomfort

Prediction: High risk of stroke

Risk Percentage: 91.87%

PREDICTION

Case 2 - Not at Risk

Age

32

Gender	Male	▼
Chest Pain	0	▼
High BP	1	▼
Short Breath	0	▼
Fatigue/We...	0	▼
Dizziness	1	▼
Edema/Sw...	0	▼
Neck/Jaw ...	0	▼
Persistent ...	0	▼
Nausea/Vo...	1	▼
Chest Disc...	0	▼
Cold Hand...	0	▼
Sleep Apnea	0	▼

Predict Stroke Risk

Prediction: Patient is NOT at Risk of Stroke
Stroke Risk Probability: 0.2082
Stroke Risk Percentage: 20.82 %

Age: 32 years
Gender: Male
Symptoms:

- High blood pressure
- Dizziness
- Nausea / Vomiting

Prediction: Low risk of stroke
Risk Percentage: 20.82%

CONCLUSION

- This study successfully compares individual and ensemble machine learning models for stroke risk prediction using clinical and symptom-based data.
- Logistic Regression achieved the best performance among individual models.
- The Ensemble Voting Classifier performed best overall, providing more balanced and stable results.
- Overall, machine learning models are effective in analyzing risk factors and enabling early prediction of stroke.

LIMITATIONS & FUTURE SCOPE

Limitations:

- Uses only structured clinical data (no real-time or imaging data)
- Trained on a single dataset (limited generalization)
- Ensemble models reduce interpretability

Future Scope:

- Include medical imaging (MRI/CT)
- Apply advanced models (Deep Learning)
- Test on larger and more diverse datasets
- Develop a web/mobile application

PUBLISHED RESEARCH PAPER

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A Comparative Study of an Individual and Ensemble Machine Learning Models for Brain Stroke Risk Prediction

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Abstract— Stroke is a problem for peoples health all around the world and it is one of the main reasons for death and long-term disability. If we can predict who is at risk of having a stroke then we can help to prevent it and get people the medical help they need on time. With more and more healthcare data available machine learning is showing a lot of promise in finding people who are at high risk of having a stroke based on their health and lifestyle.

This study focuses on comparison between individual and ensemble machine learning models for predicting brain stroke. I used a dataset that's available to the public and has information about peoples demographics and symptoms. I looked at the data to understand what it means and how different things are related to each other. I then prepared the data for use by making sure all the features are on the scale and encoding categorical features. I also made sure to leave out a feature that could give away the answer and make the models not work well.

I tried out a few machine learning models. Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. To see how well they work. We looked at how each model did by checking its accuracy, precision, recall, F1-score and ROC-AUC and we also used five-fold cross-validation to make sure the models work well in different situations. We also tried combining the strengths of selected individual models using a soft Voting Classifier. What we found out was that while Logistic Regression was really good at recalling who was at risk and had a high F1-score while the Voting Classifier that is Ensemble Model was consistently performing better at predicting who was at risk and worked well in different situations. So I chose the ensemble model as our final model. We then used this model to predict stroke risk, for patients we had not seen before. The selected model is able to give two types of results that is simple yes or no, also it gives probability of stroke risk. This study shows how machine learning models can be useful in predicting stroke risk in early stages especially with ensemble model. This can help doctors to

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